

IBM Telecom Customer Churn Analysis

Industry-Style Capstone Project Proposal Using Power BI

7043	21+	26.5%	Power BI
Customers	Features	Churn Rate	Analytics Tool

1. INTRODUCTION

Customer churn is one of the most important business challenges in the telecom industry. This project uses the IBM Telecom Customer Churn Dataset to identify patterns behind customer attrition and create interactive business intelligence dashboards.

2. DATASET INFORMATION

Source: Kaggle (IBM Telecom Customer Churn Dataset). The dataset contains 7,043 customer records with demographic information, services subscribed, billing details, contract information, payment methods, and churn status.

3. PROJECT OBJECTIVES

- Analyze customer churn behavior
- Identify high-risk customer segments
- Understand the impact of contracts and payment methods
- Study customer tenure and revenue patterns
- Build an interactive Power BI dashboard

4. TOOLS & TECHNOLOGIES

Power BI Desktop, DAX (Data Analysis Expressions), Power Query (M Language), Data Modeling, Dashboard Design and Business Intelligence Reporting.

5. PROPOSED ANALYSIS

Analysis Area	Purpose
Customer Demographics	Gender, Senior Citizen, Partner, Dependents
Contract Analysis	Compare churn across contract types
Payment Analysis	Study payment methods and churn
Service Analysis	Internet, Security, Tech Support, Streaming
Financial Analysis	Monthly Charges, Total Charges, Revenue
Retention Analysis	Tenure and churn risk identification
Executive Dashboard	KPIs and management reporting

6. DASHBOARD DESIGN

Page 1 – Executive Overview

Total Customers, Churned Customers, Churn Rate, Revenue KPIs.

Page 2 – Customer Demographics

Gender, Senior Citizen, Partner and Dependents Analysis.

Page 3 – Services & Contracts

Contract Type, Internet Service, Tech Support, Online Security.

Page 4 – Churn Insights

High-risk customer segments and revenue impact.

7. DAX MEASURES TO BE CREATED

- Total Customers
- Total Churn Customers
- Churn Rate %
- Average Monthly Charges
- Average Customer Tenure
- Revenue Metrics

8. POWER QUERY (M LANGUAGE) TASKS

- Data Cleaning
- Handling Missing Values
- Data Type Conversion
- Feature Transformation
- Data Preparation for Dashboarding

9. EXPECTED OUTCOMES

- Identify major churn drivers.
- Provide business recommendations for retention.
- Create an interactive dashboard for decision-making.
- Demonstrate practical use of Power BI, DAX, and M Language.

10. CONCLUSION

This capstone project will transform raw telecom customer data into actionable business insights through interactive Power BI dashboards. The project demonstrates Business Intelligence concepts and customer retention analytics.